INCH-POUND

MIL-N-53094(ME)
7 February 1990

MILITARY SPECIFICATION

NOZZLE ASSEMBLY, CLOSED-CIRCUIT REFUELING, ARCTIC SERVICE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a closed-circuit fuel servicing nozzle. This nozzle is intended for use with a fuel servicing adapter in a low temperature (-60 °F) environment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-P-416	- Plating, Cadmium (Electrodeposited).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-C-1752	- Cushioning Material, Packaging, Uncellular, Polyethylene Foam, Flexible.
PPP-C-1842	- Cushioning Material, Plastic, Open Cell (For Packaging Applications).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
DOD-D-1000	- Drawing, Engineering and Associated List.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-7024	- Calibrating Fluid, Aircraft Fuel System Components.
MIL-F-7179	- Finishes and Coatings: Protection of Aerospace Weapons Systems, Structures and Parts; General Specification for.
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series: General Specification for.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-S-8879	- Screw Threads, Controlled Radius Root with Increased Minor Diameter; General Specification for.
MIL-R-25988	- Rubber, Fluorosilicone Elastomer, Oil-and-Fuel Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes.
MIL-R-25988/1	- Rubber, Fluorosilicone Elastomer, Oil-and-Fuel-Resistant, O-Rings, Class 1, Grade 70.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-N-53093	- Nozzle, Adapter, Closed-Circuit to Gravity Fill, Arctic Service.
MIL-C-81740	- Coatings, Aluminum and Aluminum Alloys (Metallic Compound Decomposition).
MIL-C-81751	- Coating, Metallic-Ceramic.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
MIL-C-83413	- Connectors and Assemblies, Electrical Aircraft Grounding, General Specification for.
MIL-C-83413/7	- Connectors and Assemblies, Electrical, Aircraft Grounding: Grounding Clamp Connector For Types I and III Grounding Assemblies, Clip, Electrical.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.

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MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-2175	- Castings, Classifications and Inspection of.
MS 25384	- Electrostatic Discharger Jumper, Fuel Nozzle-to-Aircraft.
MS 27026	- Coupling Half, Quick Disconnect, Cam-Locking Type, Female, External Pipe Thread, Type VII.
MS 27029	- Coupling Half, Quick Disconnect, Cam-Locking Type, Plug, Dust, Type X.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

C13219E0482	- Strainer Body.
C13219E0484	- Strainer.
C13219E0485	- Strainer Housing.
C13219E9808	- Orifice.
C13219E9810	- Test Fixture, Closed Circuit Refueling.
C13228E1768	- Gasket, Coupling Half, Arctic Service.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 471 - Rubber Property - Effects of Liquids.
D 910 - Aviation Gasolines.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel.
D1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 31040, Miami, FL 33126.

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The closed-circuit refueling nozzle assembly as shown in figure 1 shall consist of a refueling nozzle equipped with a quick-disconnect, automatic shutoff coupler for direct connection to a vehicle fuel servicing adapter conforming to figure 2; and with an inlet fitting consisting of a 2-inch, quick-disconnect, female coupling half and fuel strainer. The nozzle assembly shall include a ground cable assembly, a dust plug for the inlet fitting, and a dust plug or cap for the coupler. The nozzle assembly can also be adapted to perform conventional open port or gravity fill dispensing by attaching an adapter nozzle conforming to MIL-N-53093.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the nozzle. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The use of magnesium or any alloy thereof is prohibited.

3.3.1 Material deterioration prevention and control. The nozzle shall be fabricated from compatible materials, inherently corrosion resistant or treated

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to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the nozzle may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Castings. Castings shall be clean, sound, and free from blowholes, porosity, cracks and other defects. Castings shall conform to MIL-STD-2175, class suitable for the application.

3.3.5 Threaded parts. All screw threads shall be in accordance with MIL-S-7742, MIL-S-8879, or FED-STD-H28.

3.3.6 Finish and protective coating. The system shall be finished with applicable materials and corrosion protective systems selected in accordance with MIL-F-7179, type II and as specified below. When required to provide a path for electrical current, protective coatings may be removed locally.

3.3.6.1 Cadmium plating. Cadmium plating shall conform to QQ-P-416, class 1, type II. Cadmium plating shall not be used on surfaces which would normally be in contact with fuel. In lieu of cadmium plating, the parts can be aluminum coated conforming to MIL-C-81740, class 1, type II; or metallic-ceramic coated conforming to MIL-C-81751, type II.

3.3.6.2 Chromium plating. Chromium plating shall conform to QQ-C-320, class 1, optional finish.

3.3.6.3 Anodizing. Aluminum alloys shall be protected to resist corrosion and wear in accordance with MIL-A-8625, type II, class 1 (minimum thickness 0.0007-inch or 18 micrometers).

3.3.6.4 Chemical films. Aluminum alloys which do not anodize satisfactorily shall be chemically coated in accordance with MIL-C-5541, class 1A. When abrasion resistance is a factor, chemical films shall not be used in lieu of anodizing.

3.3.6.5 Treatment and painting. The portions of the nozzle normally painted, internally and externally, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Mating surfaces and surfaces in contact with fuel shall not be painted. Top coat color shall be camouflage green 383, color no. 34094, type II conforming to MIL-C-46168.

3.3.7 Sealing compound. Thread sealing compound shall be applied to all mating pipe threads prior to assembly.

3.3.8 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and

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reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the nozzles may be newly fabricated from recovered materials to the maximum extent practicable, provided the nozzles produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the nozzles.

3.3.9 Standard parts. Standard parts (MS, AN, or NAS) shall be used wherever they are suitable for the purpose and shall be identified in all parts lists by the applicable military PIN number. In the event suitable standard parts are not available, commercial parts may be used provided they can be identified by the vendor's part number and can be used without modification.

3.4 Environmental.

3.4.1 Operating conditions. The nozzle shall perform as specified herein during or after exposure to the following environments:

a. Temperature range: -60 to +95 °F.
b. Relative humidity: Zero to 100 percent.
c. Vibration: Vibration incident to service.
d. Salt spray: Exposure to salt sea atmosphere.
e. Sand and dust: Exposure to dusty arctic environments.
f. Shock: No damage when tested as specified in 4.5.2.15.

3.4.2 Storage. The nozzle shall not be damaged by storage in all ambient temperatures between -65 and +145 °F.

3.5 Human factors engineering. The nozzle shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited, to general requirements (4); controls (5.4); labeling (5.5); anthropometry (5.6); design for maintainability (5.9); and hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.6 Maintainability.

3.6.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, lubrication charts, technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manuals as specified (see 6.4), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.18.

3.6.2 Scheduled maintenance. The time required for the operator to accomplish all scheduled preventative maintenance shall not exceed 5 minutes daily.

3.7 Closed-circuit refueling nozzle. The nozzle shall be equipped with a coupler for direct connection to a closed-circuit adapter conforming to MIL-N-53093, an internal pressure regulation device, a manually controlled on-off flow valve, and a ground cable assembly. The nozzle shall be designed so that one person wearing arctic mittens can easily couple it to a vehicle adapter. The nozzle shall be suitable for operation under all possible operating

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conditions from 0 to 115 gallons per minute (gpm) flow rate and 0 to 125 pounds per square inch (psi) nozzle inlet pressure. The nozzle shall withstand an internal pressure of 225 psi without damage.

3.7.1 Coupler. The nozzle shall include a quick-disconnect, automatic-shutoff coupler suitable for connecting the nozzle to any adapter conforming to figure 2. The coupler shall automatically engage the adapter with a mechanical lock which eliminates the need for the operator to maintain any continuous manual hold force. When the nozzle is not coupled to an adapter, a seal which functions automatically shall make it impossible to initiate or maintain flow through the nozzle. Fuel spillage which may occur when the nozzle is uncoupled from the adapter shall not exceed 30 cubic centimeters (cc) when the fuel flow control valve is closed or 150 cc when flowing at the maximum flow rate. The coupler shall include a dust plug or cap to prevent dust from entering that portion of the nozzle which engages the adapter. The dust plug or cap shall be adequately secured to the nozzle to prevent loss and shall withstand a pull test of 75 pounds.

3.7.2 Pressure regulation. The nozzle shall include an integral means to regulate the pressure at the interface between the nozzle and a closed-circuit adapter. The pressure regulator shall maintain the interface pressure of 15 pounds per square inch gauge (psig) nominally for nozzle inlet pressures from 20 to 125 psi. When the nozzle is connected to a closed-circuit adapter discharging through a fixed orifice, the pressure regulator shall function as a means of flow regulation. The regulated flow shall be within the limits required when the nozzle is tested as specified in 4.5.2.4. When the nozzle inlet pressure is less than that required to produce an interface pressure of at least 15 psig, the pressure regulating device shall remain in the full open position. With the nozzle connected to a closed-circuit adapter and the flow control valve in the open position, it shall be possible to visually determine when the regulator has shut off fuel flow.

3.7.3 Flow control valve. The nozzle shall include a flow control valve which allows manual control of the flow rate from no flow to full flow. The flow control valve shall remain in the full closed or full open position as selected by the operator without the need of maintaining a continuous manual holding force. The valve must have identification markings as specified in 3.11.2.

3.7.4 Nozzle inlet. The inlet of the nozzle body shall be a 1-1/2-inch by 11-1/2 NPT internal thread. The centerline of the nozzle inlet shall be located 120 ±10 degrees from the centerline of the nozzle outlet as shown in figure 1.

3.7.5 Ground cable assembly. The nozzle assembly shall be equipped with a ground cable assembly for grounding the nozzle to the vehicle and to a ground stake. The nozzle-to-vehicle ground cable shall conform to MS 25384-1, except the connector shall conform to MIL-C-83413/7. The nozzle-to-ground stake cable shall be coated wire 15 feet long, and the end shall be equipped with a clamp in accordance with MIL-C-83413/7, suitable for bonding the cable assembly to a 5/8-inch diameter ground stake.

3.7.6 Disengagement. The force required to remove the nozzle from the closed-circuit adapter shall be not more than 32 pounds nor less than 18 pounds.

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3.8 Fuel resistance. The nozzle shall be resistant to, and suitable for use with, fuels and fluids conforming to MIL-G-3056, MIL-T-5624, MIL-C-7024, MIL-T-83133, ASTM D 910, and ASTM D 471, reference fuel B.

3.9 Physical characteristics.

3.9.1 Dimensions. The nozzle, including all actuating handles, shall be contained within the envelope shown on figure 1. Dimension of the nozzle shall be appropriate for use by personnel wearing arctic mittens, in accordance with 3.5.

3.9.2 Strength. The nozzle shall withstand, while operating, a side load of sufficient magnitude to produce a moment of 100 foot-pounds about the intersection of the nozzle outlet centerline and the locking lug reference plane on a closed-circuit adapter (see figure 3).

3.9.3 Electrical conductivity. When the nozzle is coupled to a closed circuit adapter, the resistance between the adapter and the plug on the ground cable shall not exceed 10 ohms.

3.9.4 Weight. The weight of the nozzle shall be not more than 8 pounds.

3.10 Lubrication. The nozzle shall operate satisfactorily without the use of lubricants other than solid film lubricants, and that provided by fuel being dispensed through the nozzle.

3.11 Identification marking.

3.11.1 Identification. The nozzle shall be identified by rolling, stamping, or stenciling in accordance with MIL-STD-130.

3.11.2 Fuel flow marking. The flow control valve operating device shall be provided with double ended arrows showing the direction of operations and labeled at each end to indicate the functional result ("open", "close", etc).

3.12 Workmanship.

3.12.1 Castings and forgings. All parts, components, and assemblies of the nozzle which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect functions.

3.12.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.12.3 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter other than mill scale that can be

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removed by chipping and wire brushing. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the nozzle to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16-inch, unless appropriate bridging techniques are used.

3.12.4 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code - Steel.
AWS D1.2 - Structural Welding Code - Aluminum.
ASME Boiler and Pressure Vessel Code, Section IX.

Contractors who only make horizontal welds need not qualify welders for "all position welding". In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. The test results shall be made available, upon request, for review by the contracting officer or the contracting officer's representative.

3.12.5 Welding practices.

a. Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's recommendations.
b. Tack welds shall be subject to the same quality requirements as final welds.
c. Work shall be positioned for flat welding whenever practicable.
d. Procedure and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and nozzle.

3.12.6 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to insure that when the welded parts are assembled together to make the nozzle, the nozzle shall conform to the inspection requirements specified in section 4.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to both the appearance and strength of the weldment.
c. Undercut in weldments shall be held to a minimum but shall not be more than 1/32-inch deep or 5 percent of the thickness of the parent metal, whichever is smaller.

3.12.7 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where

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tion. The nozzle shall be operated for [illegible] cycles

necessary. Matching thread areas securing bolts shall be sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.12.8 Machine work. Tolerances and gauges for metal fits shall conform to the limits specified herein and on applicable drawings and to the standards of the industry.

3.13 Interchangeability. All parts having the same vendor part number shall be directly and completely interchangeable with respect to installation and performance. Changes in vendor part numbers shall be governed by drawing number requirements of DOD-D-1000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Disassembly inspection. Failure of any examination or test of the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, and drawings.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

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in a single container. preserved and packaged

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).
d. Inspection comparison (see 4.7).

4.3 First article inspection. Unless otherwise specified (see 6.2), two first article nozzle assemblies shall be examined as specified in 4.5.1 and subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection. Unless otherwise specified (see 6.2), the Government reserves the right to perform first article testing on three or more nozzle assemblies. The tests shall consist of any of the tests of table I which the Government desires to perform.

4.3.1 Rejection and retest. When a nozzle fails to meet any of the requirements of the first article tests, the cause of failure shall be determined and the disassembly inspection performed as specified in 4.1.2. Full particulars of the failure and the action required to correct the fault shall be reported to the contracting officer. All first article tests which are affected by the corrections shall be repeated.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examinations and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found, and rejected when one or more defects are found.

4.4.2 Examination. Sample selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each nozzle shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table I. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The nozzle shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be corrosion resistant in the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatments.
105. Castings have holes, are not clean, or not as specified.
106. Fabrication not in accordance with drawings.

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107. Parts or components missing or not as specified.
108. Assembly incomplete or incorrect.
109. Finish and protection coating not as specified.
110. Cadmium plating not as specified.
111. Chromium plating not as specified.
112. Anodizing not as specified.
113. Chemical films not as specified.
114. Treatment and painting not as specified.
115. Paint color not as specified.
116. Sealing compound not as specified.
117. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the nozzles.
118. Human engineering not as specified.
119. Fuel flow valves and markings not as specified.
120. Ground cable assembly not as specified.
121. Dimensions not as specified.
122. Weight not as specified.
123. Lubrication not as specified.
124. Identification or special marking missing or illegible.
125. Workmanship not as specified.
126. Welding not as specified.
127. Welding practices not as specified.
128. Welder/operator certification not as specified.
129. Interchangeable parts or components not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests may be performed without shelter and at the climatic conditions existing at the place of test. The nozzles shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

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TABLE I. Test schedule.

First article	Quality conformance		Test	Test paragraph	Requirement paragraph
	Individual	Samples			
1	2	3	4	5	6
X	X	X	Test conditions.	4.5.2.1	-
X	X	X	Functional test.	4.5.2.2	3.7
X	-	X	Strength.	4.5.2.3	3.7 and 3.9.2
X	-	X	Regulator.	4.5.2.4	3.7.2
X	X	X	Electrical resistance.	4.5.2.5	3.9.3
X	-	X	Coupler spillage, normal.	4.5.2.6	3.7.1
X	-	X	Coupler spillage, under pressure.	4.5.2.7	3.7.1
X	-	X	Coupler spillage, flowing.	4.5.2.8	3.7.1
X	-	-	Fuel resistance.	4.5.2.9	3.8
X	-	-	Low temperature.	4.5.2.10.1	3.4.2, 3.4.2 and 3.9.1
X	-	-	High temperature.	4.5.2.10.2	3.4.1, 3.4.2
X	-	-	Vibration.	4.5.2.11	3.4.1
X	-	-	Sand and dust.	4.5.2.12	3.4.1
X	-	-	Humidity.	4.5.2.13	3.4.1
X	-	-	Accelerated corrosion.	4.5.2.14	3.4.1
X	-	-	Performance.	4.5.2.17	Section 3
X	-	-	Drop test.	4.5.2.15	3.4.1
X	X	-	Burst pressure.	4.5.2.16	3.7
X	-	-	Disassembly inspection.	4.1.2	Drawings
X	-	X	Disengagement.	4.5.2.19	3.7.6

4.5.2.1.1 Cleaning. Before testing any nozzle, all oil, grease, or any other corrosion-resistant compound shall be removed from the interior and exterior surfaces.

4.5.2.1.2 Test fluid. Except as specified herein, all tests shall be conducted using a test fluid conforming to MIL-T-5624, grade JP-4; MIL-C-7024, type II. The test fluid temperature shall be between -60 and +95 °F, with the exception of the low temperature test 4.5.2.10.1 and high temperature test 4.5.2.10.2. All flow rate performance data shall be corrected to a specific gravity of 0.78.

4.5.2.2 Functional test. With the nozzle flow control valve closed, apply a pressure of 25 psi to the nozzle inlet. Connect the nozzle to a closed-circuit fuel servicing adapter, hereinafter referred to as a test adapter, conforming to figure 2. Open the nozzle flow control valve initiating flow. Block the test adapter discharge port and check for visual indication that the regulator has shut

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off flow. Close the nozzle flow control valve and disconnect the nozzle from the test adapter. Open the test adapter discharge port, increase the nozzle inlet pressure to 125 psi and connect the nozzle to the test adapter. Open the nozzle flow valve and adjust flow rate to maintain 125 psi nozzle inlet pressure. Close the nozzle flow control valve and disconnect the nozzle from the test adapter. Any evidence of leakage, binding, valve chatter, distortion, or requiring excessive operating force shall constitute failure of this test.

4.5.2.3 Strength.

4.5.2.3.1 Proof pressure-open. Connect the nozzle to a test adapter with the test adapter port blocked. With the nozzle flow control valve in the open position, apply a pressure of 5 psi to the nozzle inlet for a period of not less than 1 minute. Repeat this procedure increasing the nozzle inlet pressure in 25-psi increments until an inlet pressure of 180 psi is reached. Any evidence of leakage or distortion shall constitute failure of this test.

4.5.2.3.2 Proof pressure-closed. With the nozzle dust cap removed and the flow control valve in the closed position, pressure shall be applied to the nozzle inlet in accordance with 4.5.2.3.1. If the nozzle flow control valve can be put in flow position without connecting the nozzle to an adapter, the test procedure shall be repeated with the flow control valve in the full open position. Any evidence of leakage or distortion shall constitute failure of this test.

4.5.2.3.3 Side load. With the nozzle connected to a rigidly mounted test adapter with the discharge port blocked, apply 125 psi pressure to the nozzle inlet. Apply a load to the nozzle inlet of such magnitude to cause a moment of 100 foot-pounds about the intersection of the nozzle outlet centerline and the locking lug reference plane on the test adapter as shown in figure 3. Actuate the flow control mechanism from the closed, to the open, to the closed positions. Any evidence of leakage, binding or distortion shall constitute failure of this test.

4.5.2.3.4 Pull test. All dust cap and plug security chains and cables shall be connected to the nozzle and subjected to a proof load of 75 pounds. Any breakage shall constitute failure of this test.

4.5.2.4 Regulator. Connect the nozzle to a test fixture conforming to Drawing C13219E9810. With orifice plate 13219E9808-1 installed in the test fixture, slowly increase the nozzle inlet pressure from zero to 125 psi. The flow rate through the nozzle shall be measured for nozzle inlet pressures of 25, 50, 75, 100, and 125 psi. Replace the test fixture orifice plate and repeat the test for each orifice in table II. Any evidence of valve chatter or unstable flow conditions at any nozzle inlet pressure from zero to 125 psi shall constitute failure of this test. Failure of the nozzle to regulate flow within the limits specified in table II for nozzle inlet pressures from 25 to 125 psi shall constitute failure of this test. Any evidence of leakage shall also constitute failure of this test.

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TABLE II. Regulator test limits.

Orifice	Flow Rate (gpm)		
	Minimum	Normal	Maximum
13219E9808-1	17.0	20	23.0
13219E9808-2	59.0	65	70.0
13219E9808-3	85.0	100	115.0

4.5.2.5 Electrical resistance. With the nozzle connected to a test adapter, an ohmmeter shall be used to measure the resistance between the test adapter and the plug on the ground cable assembly. Resistance exceeding 10 ohms shall constitute failure of this test.

4.5.2.6 Coupler spillage, normal. The nozzle shall be coupled to a test adapter with the discharge port open. With a flow rate of not less than 50 gallons per minute (gpm) through the nozzle, the discharge port on the test adapter shall be blocked. The nozzle flow control valve shall be in the closed position and the nozzle uncoupled from the test adapter. Spillage of test fluid from between the nozzle and test adapter exceeding 30 cc shall constitute failure of this test.

4.5.2.7 Coupler spillage, under pressure. If the coupler and nozzle flow control valve are actuated by separate mechanisms, the test as described in 4.5.2.6 shall be repeated, except the nozzle flow control shall be in the full open position when the nozzle is uncoupled from the test adapter. Spillage of test fluid from the nozzle exceeding 150 cc shall constitute failure of this test.

4.5.2.8 Coupler spillage, flowing. If the coupler and nozzle flow control valve are actuated by separate mechanisms, the nozzle shall be tested as follows. When flowing at a rate of not less than 100 gpm the nozzle shall be uncoupled from the test adapter. The nozzle flow control valve shall be in the full open position during the uncoupling operation. Spillage of test fluid from the nozzle exceeding 150 cc shall constitute failure of this test.

4.5.2.9 Fuel resistance. The test specified in 4.5.2.9.1 through 4.5.2.10 shall be conducted in the sequence listed and each test shall follow immediately after the preceding test.

4.5.2.9.1 Fuel soak, phase I. Immerse the nozzle in test fluid conforming to ASTM D, reference fuel B, at a temperature of +95 ±5 °F for a period of 14 days. During the first 7 days, the control valve shall be in the closed position. During the last 7 days, the flow control valve shall be in the open position. After a minimum of 14 days fuel soak, the nozzle shall be tested as specified in 4.5.2.2 and 4.5.2.4.

4.5.2.9.2 Drying cycle, phase I. Drain all test fluid from the nozzle and allow it to dry for 24 hours in an ambient air temperature of +95 ±5 °F. The nozzle then shall be tested as specified in 4.5.2.2 and 4.5.2.4.

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4.5.2.9.3 Fuel soak, phase II. Repeat 4.5.2.9.1, except fuel soak period shall be 7 days minimum with the flow control valve in the open position for 3 days, then in the closed position for 4 days.

4.5.2.9.4 Drying cycle, phase II. Repeat 4.5.2.9.2 except the drying period shall be 30 hours.

4.5.2.10 Environmental.

4.5.2.10.1 Low temperature. The nozzle shall be tested in accordance with MIL-STD-810, method 502.2, procedures I, II, and III. Immerse the nozzle in test fluid conforming to ASTM D 471, reference fuel B at a temperature of -60 °F maximum for a period of 48 hours. After the 48-hour soak period, and with the ambient and test fluids at a temperature no higher than -60 °F, test the nozzle as specified in 4.5.2.2, 4.5.2.3 and 4.5.2.4 and for 250 cycles each at 15, 50, and 125 psig. A cycle is defined in 4.5.2.17. Nonconformance to 3.4 and 3.9.1 shall constitute failure of this test.

4.5.2.10.2 High temperature. The nozzle shall be tested in accordance with MIL-STD-810, method 501.2, procedures I and II. Immerse the nozzle in test fluid conforming to ASTM D 471, reference fuel B at a temperature of +145 °F for a period of 48 hours. After the 48-hour soak period, and with the ambient and test fluids at a temperature no higher than +95 °F, test the nozzle as specified in 4.5.2.2, 4.5.2.3 and 4.5.2.4 and for 250 cycles each at 15, 50, and 125 psig. A cycle is defined in 4.5.2.17. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.11 Vibration test. The nozzle shall be tested in accordance with part 2, procedure XI, method 514.2 of MIL-STD-810. After being subjected to the specified vibration, the nozzle shall be examined and tested in accordance with 4.5.2.2, 4.5.2.3, and 4.5.2.4.

4.5.2.12 Sand and dust. The nozzle shall be tested in accordance with method 510.1 of MIL-STD-810. Upon completion of the test, the nozzle shall be tested in accordance with 4.5.2.2 and 4.5.2.4. Cleaning of the nozzle prior to testing is not permissible.

4.5.2.13 Humidity. The nozzle shall be tested in accordance with procedure I, method 507.1 of MIL-STD-810. Upon completion of the humidity testing, the nozzle shall be tested as specified in 4.5.2.2 and 4.5.2.4.

4.5.2.14 Accelerated corrosion. The nozzle shall be immersed in a solution of 2.5 percent by weight NaCl in distilled water. After immersion, the solution shall be drained and the nozzle shall be heated in an oven to a temperature of 130 ±5 °F for a period of not less than 1 hour. The immersion and heating cycle shall be repeated 50 times. The nozzle shall not be operated at any time during the immersion cycles. Immediately after completing the immersion cycles, the nozzle shall be washed out with warm water to remove salt accumulation. The nozzle shall be dried; wet with fuel; and tested as specified in 4.5.2.2, 4.5.2.3, and 4.5.2.4. Corrosion of any part of the nozzle to a degree that affects performance shall constitute failure of this test.

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4.5.2.15 Drop test. With the nozzle attached to a 10-foot length of 1.5-inch diameter hose, the dust cap removed, and the upstream portion of the hose laying at floor level, the nozzle shall be dropped on a concrete floor from a height of 6 feet. The nozzle shall be dropped five times with at least one impact occurring on the nose plane and one impact on the operating handle. The remainder of the impacts shall occur on that part of the nozzle considered to be most vulnerable to damage. Upon conclusion of the drops, the nozzle shall be tested as specified in 4.5.2.2, 4.5.2.3, and 4.5.2.4.

4.5.2.16 Burst pressure. With the nozzle flow control valve in closed position, apply a hydrostatic pressure of 250 psi to the nozzle inlet and maintain the pressure for not less than 3 minutes. Release the pressure; connect the nozzle to a test adapter with the discharge port blocked and repeat the pressure test. Any evidence of leakage or distortion shall constitute failure of this test.

4.5.2.17 Performance. The nozzle shall be subjected to the series of operational cycles in accordance with phases I, II, III, IV, V, and VI, as shown in table III. For phases IV and V, the test fluid shall contain contaminants in accordance with table IV. Upon completion of each phase, the nozzle shall be tested in accordance with 4.5.2.4. One test cycle shall consist of the following:

a. Connect nozzle to a test adapter. (The discharge port of the test adapter shall be open for flow.)
b. Position the nozzle flow valve in the open position.
c. Shut off flow by blocking the discharge port of the test adapter after establishing full flow for approximately 10 seconds.
d. Position the nozzle flow control valve in the closed position.
e. Disconnect nozzle from the test adapter.

Any evidence of fuel leakage or inability of the nozzle to pass the test specified in 4.5.2.4 and 4.5.2.17 at the conclusion of each test phase shall constitute failure of this test.

TABLE III. Endurance test.

Phase	Cycles	Test fluid	Nozzle inlet pressure
I	500	Clean	15 psig
II	500	Clean	50 psig
III	500	Clean	125 psig
IV	500	Contaminated	15 psig
V	500	Contaminated	125 psig
VI	500	Clean	50 psig

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TABLE IV. Fuel contaminants.

Contaminant	Particle size	Quantity (minimum)
Iron oxide.	0 - 5 microns 5 - 0 microns	28.5 gm/400 gal. 1.5 gm/400 gal.
Sharp silica sand.	150 - 300 microns 300 - 420 microns	1.0 gm/400 gal. 1.0 gm/400 gal.
Prepared dirt conforming to AC Spark Plug Co. Part No. 1543637 or equal (coarse Arizona road dust).	Mixture as follows: 0 - 5 microns (12%) 5 - 10 microns (12%) 10 - 20 microns (14%) 20 - 40 microns (23%) 80 - 200 microns (9%)	8.0 gm/400 gal.
Cotton linters.	Staple below 7 (U.S. Department of Agriculture Grading Standards)	0.1 gm/400 gal.
Crude naphthenic acid Salt water solution shall contain 4 parts NaCl to 96 parts H_2O by weight.		0.03% by volume 0.01% entrained

4.5.2.18 Maintainability. Maintenance operations required during all testing shall be accomplished by operating and support personnel in accordance with the requirements of 3.6.1 and 3.6.2. Failure of the support elements to maintain the nozzles as specified shall constitute failure of this test.

4.5.2.19 Disengagement. Follow procedures in items a., b., c., and d. of 4.5.2.17. The nozzle shall then be disengaged from the test adapter by applying a steady pull along the axis of the nozzle discharge barrel. The procedures of a., b., c., and d., and disengagement shall be repeated 10 times. The force required for disengagement shall be measured to the nearest pound with a calibrated force gauge. Failure to meet the requirement of 3.7.6 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection.

4.6.1.1 Unit of product. The unit of product shall be a quantity of nozzles preserved and packaged in a single container.

4.6.1.2 Sampling. Sampling shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found, and rejected when one or more defects are found.

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4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects:

130.	Nozzles not preserved as specified for level A or C (see 5.1.1 and 5.1.2).
131.	Shipping containers not as specified for levels A and C (see 5.2.1 and 5.2.2).
132.	Shipping container weight and size exceeding specified limitations for levels A and C (see 5.2.1 and 5.2.2).
133.	Strapping not as specified for level A (see 5.2.1).
134.	Marking incorrect, missing or illegible (see 5.3).

4.7 Inspection comparison. The Government may select nozzles at any time during the contract production period and subject these nozzles to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article nozzles. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of any nozzle to meet any requirement specified herein, during, and as a result of the examination and tests specified in 4.7, shall be cause for rejection of the nozzle and shall be cause for refusal by the Government to continue acceptance of production nozzles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on nozzles produced and accepted under the contract. Any deficiencies found as a result of comparison inspection will be considered prima facie evidence that all nozzles accepted prior to completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each nozzle shall be preserved in accordance with the method III requirements of MIL-P-116, utilizing a fiberboard carton as the unit container; the carton shall conform with PPP-B-636, class weather resistant, grade W6c. Cushioning material conforming to PPP-C-1752 or PPP-C-1842 shall be used, as necessary.

5.1.2 Level C. Each nozzle shall be preserved as specified for level A, except the boxes shall be class domestic.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Nozzles, preserved as specified in 5.1 shall be packed in boxes conforming with PPP-B-601, overseas type, style optional. The gross weight

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of the shipping container and contents shall not exceed 150 pounds. Boxes shall be strapped in accordance with the appendix to the box specification except strapping shall be flat steel and shall be finish B.

5.2.2 Level C. Nozzles, preserved as specified in 5.1, shall be packed as specified for level A, except the boxes shall be domestic type and the strapping may be finish A.

5.3 Marking. Marking of both unit and shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nozzle covered by this specification is intended for use with fuel dispensing systems to safely deliver contaminant-free fuel to military aircraft, vehicles, and other powered equipment in a low temperature (-60 °F).

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is not required for inspection and approval (see 3.2 and 4.3).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of the nozzles.

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6.5 Subject term (key word) listing.

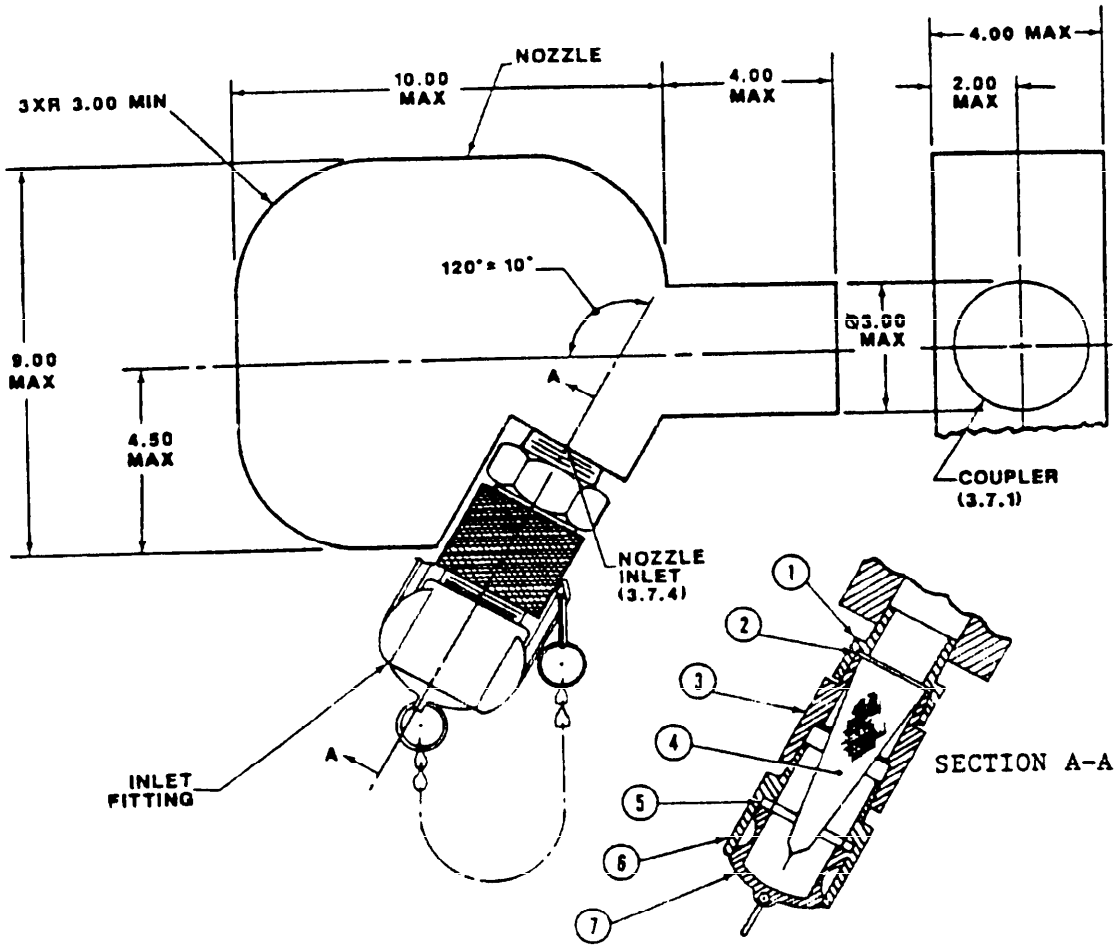
Adapter, fuel servicing
Arctic
Assembly, ground cable
Cap
Castings
Coupler
Dust plug
Fluids
Fuel, powered equipment
Fuel, vehicles
Fuel, aircraft
Nozzle
Plating, cadmium
Plating, chromium
Regulator, pressure
System, fuel dispensing
Valve, flow control

Custodian:
Army - ME

Preparing activity:
Army - ME

(Project 4930-A270)

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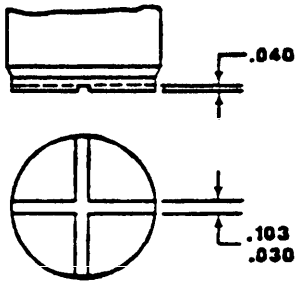


ITEM NO.	PART NO.	DESCRIPTION
1	13219E0485	STRAINER HOUSING
2	M25988/1-226	O-RING, MIL-R-25988/1
3	13219E0482	STRAINER BODY
4	13219E0484	STRAINER
5	13228E1768-6	GASKET, COUPLING HALF, 2"
6	MS27026-11	COUPLING HALF, FEMALE, 2"
7	MS27029-11	COUPLING HALF, PLUG, 2"

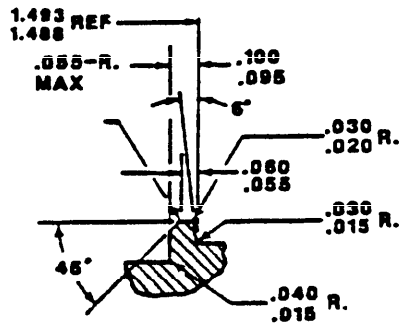
FIGURE 1. Nozzle assembly.

X-4839

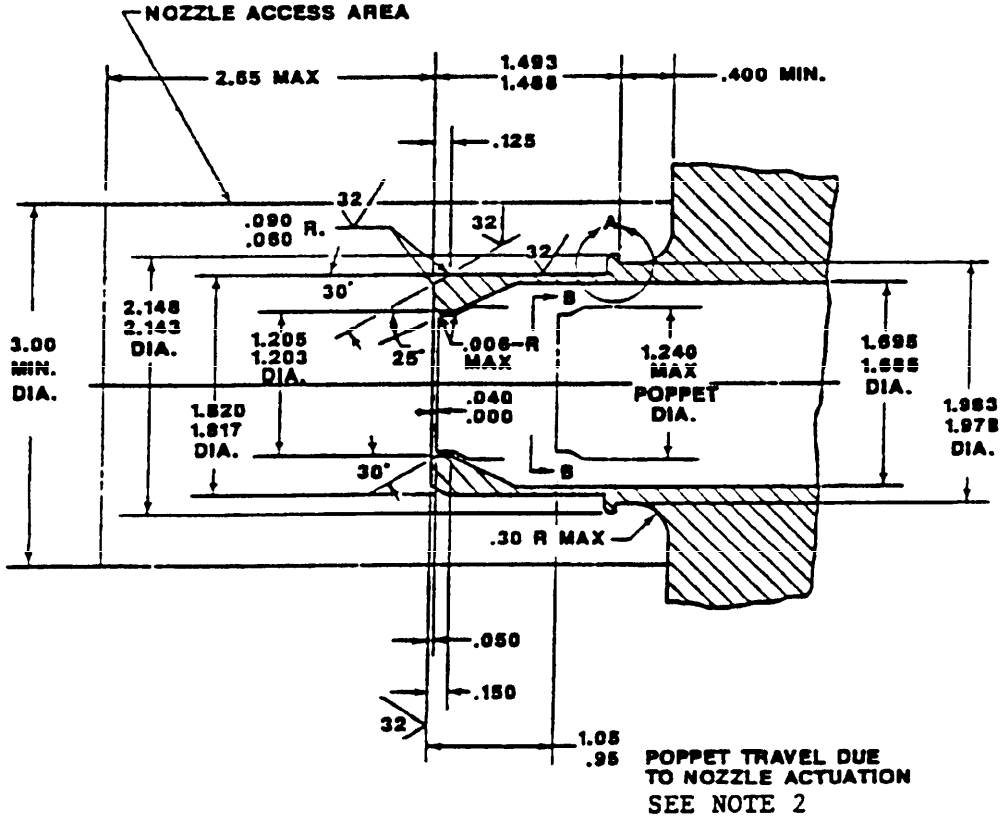
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VIEW B-B



VIEW A



NOTES:

1. TOLERANCES, UNLESS OTHERWISE SPECIFIED: DECIMALS ±.010, ANGLES ±0°30'.
2. POPPET SHALL BE SELF-SEALING WITH A SPRING LOAD OF 3 ±.3 LB SEATED, 3 LB/INCH MAXIMUM SPRING RATE.

FIGURE 2. Adapter configuration, closed-circuit fuel servicing.

X-4840

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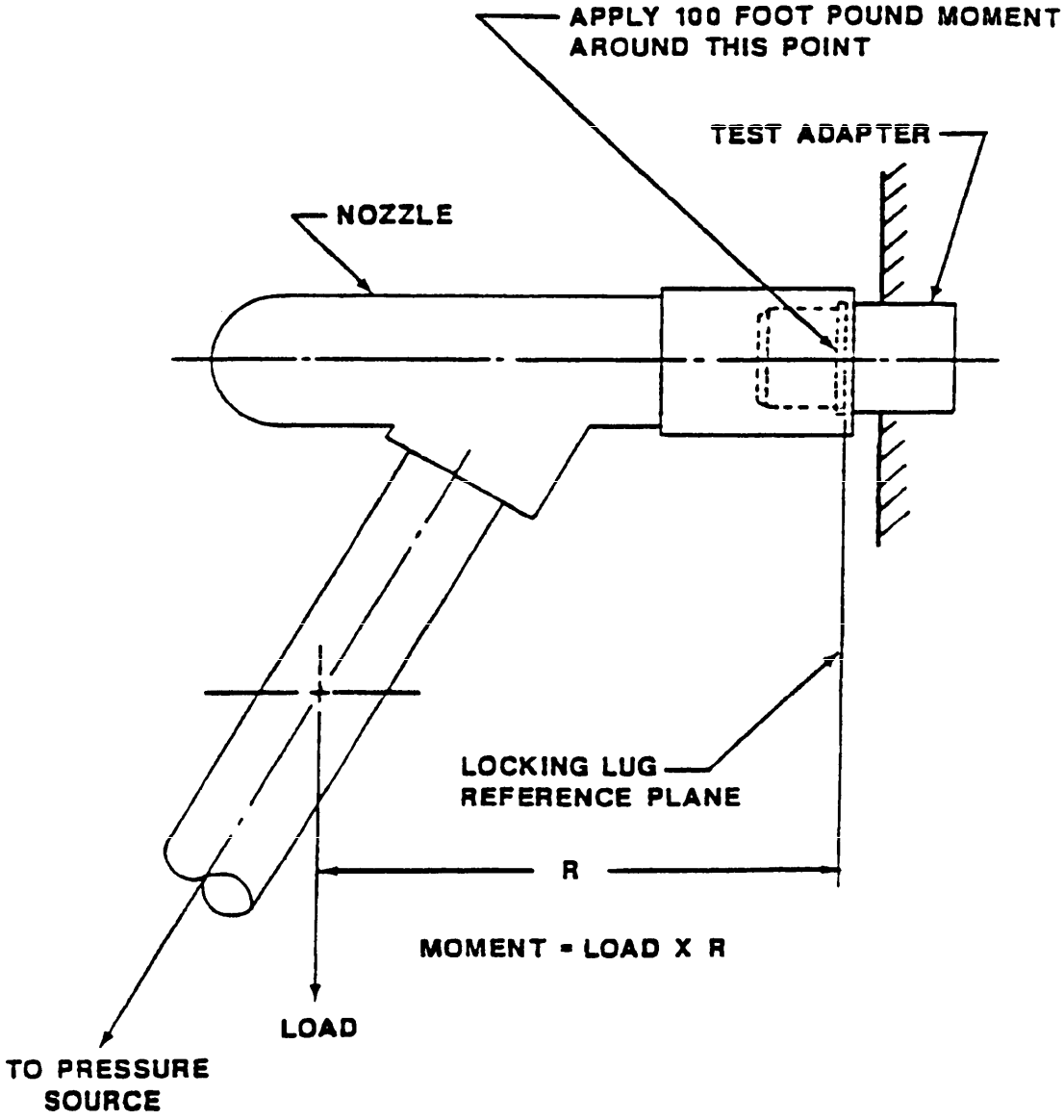


FIGURE 3. Side load test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-N-53094 (ME)	900207

3. DOCUMENT TITLE
Nozzle Assembly, Closed-Circuit Refueling, Arctic Service

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. MAILING ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
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US Army Belvoir RDE Center
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